INCH POUND

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PERFORMANCE SPECIFICATION

FOAM, RIGID, STRUCTURAL, CLOSED CELL

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers rigid, structural, closed cell foam (see 6.1).

1.2 Classification. The foam is to be of the following types and classes:

1.2.1 Type I. Type I foams are processed at temperatures up 250°F (121°C) under the following classes organized by density.

Class 1 at 2.0 lb/ft^3 (32.0 kg/m^3) density, nominal.
Class 2 at 3.2 lb/ft^3 (51.3 kg/m^3) density, nominal.
Class 3 at 4.7 lb/ft^3 (75.3 kg/m^3) density, nominal.

1.2.2 Type II. Type II foams are processed at temperatures greater than 250°F (121°C) and up to 350°F (177°C) after post-curing under the following classes organized by density (see 6.4).

Class 2 at 3.2 lb/ft^3 (51.3 kg/m^3) density, nominal.
Class 3 at 4.7 lb/ft^3 (75.3 kg/m^3) density, nominal.
Class 4 at 6.9 lb/ft^3 (110.5 kg/m^3) density, nominal.
Class 5 at 12.8 lb/ft^3 (205.0 kg/m^3) density, nominal.
Class 6 at 18.7 lb/ft^3 (299.5 kg/m^3) density, nominal.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

ASTM INTERNATIONAL

ASTM C273/273M	-	Standard Test Method for Shear Properties of Sandwich Core Materials
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D910	-	Standard Specification for Aviation Gasolines
ASTM D1621	-	Standard Test Method for Compressive Properties of Rigid Cellular Plastics
ASTM D1622	-	Standard Test Method for Apparent Density of Rigid Cellular Plastics

(Copies of these documents are available online at https://www.astm.org/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

3.3 Material. The material shall be a dimensionally stable, closed cell, rigid foam. The foam material shall be homogeneous throughout. The cells shall be of approximately the same size, without large accumulations of unexpanded resin, or other inclusions. The foam shall be free of cracks, surface irregularities, handling damage, and contaminants. The foam shall conform to the dimensional requirements of the purchase order.

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3.4 Thermal stability. Type I expanded foam shall maintain dimensional stability at temperatures up to 250°F (121°C). Type II foam shall maintain dimensional stability at temperatures up to 350°F (177°C).

3.5 Marking. Each individual sheet shall be permanently marked in one corner with the following information:

a. Material specification number, revision letter

b. Type and class

c. Lot number and batch number

d. Date of manufacture

e. Manufacturer's product designation

3.6 Properties. Unless otherwise specified the physical and mechanical properties of the foam shall be as specified herein when tested in accordance with the test methods listed in Table V.

3.7 Physical properties. The density range and maximal water absorption of the foam shall be as specified in Table I.

TABLE I. Physical properties of rigid foam.

Class	Density		Water Absorption (%), Max [1/]	
	lb/ft^3	kg/m3	Type I	Type II
1	1.56 to 2.44	(25.0 - 39.1)	13	---
2	2.45 to 3.85	(39.2 - 61.7)	13	10
3	3.86 to 5.59	(61.8 - 89.5)	13	10
4	5.60 to 8.20	(89.7 - 131.3)	---	10
5	10.60 to 15.00	(169.8 - 240.3)	---	10
6	15.60 to 21.80	(249.9 - 349.2)	---	10

[1/] The procedure in paragraph 4.5 shall be used to determine water absorption.

3.8 Fluid resistance. The foam shall be resistant to the fluids listed in Table II. The compressive strength of the foam at 73.4 ± 3.6°F (23.0 ± 2.0°C) shall remain above the minimal values specified in 3.3.3 when immersed in the fluids for the exposure times and temperatures as specified in Table II.

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TABLE II. Fluid resistance times and temperatures of immersion. [1/]

Fluid	Immersion Time	Temperature
Gasoline (ASTM D910, octane 100)	1500 hours	73.4±3.6°F (23±2.0°C)
Hydraulic oil, petroleum base	1500 hours	73.4±3.6°F (23±2.0°C)
Hydraulic oil, synthetic hydrocarbon base	1500 hours	73.4±3.6°F (23±2.0°C)
JP-4 turbine fuel, aviation	1500 hours	73.4±3.6°F (23±2.0°C)
Lubricating oil, aircraft turbine engine, synthetic base with a kinematic viscosity of 4.0 ± 1.0 centistokes [4.3 ± 1.1(10)$^{-5}$ ft^2/sec] at 212°F (100°C)	1500 hours	73.4±3.6°F (23±2.0°C)
Deicing fluid (propylene glycol base with a specific gravity of 1.103 ± 0.003 or ethylene glycol base with a specific gravity of 1.105 ± 0.005)	1500 hours	73.4±3.6°F (23±2.0°C)
Acetone (If the end item will be exposed to methyl ethyl ketone, MEK shall replace acetone)	1500 hours	73.4±3.6°F (23±2.0°C)
Fire resistant phosphate ester hydraulic fluid (SAE AS 1241, type IV, class I)	1500 hours	158.0±3.6°F (70±2.0°C)

[1/] Specimens shall be dried prior to immersion and prior to testing in accordance with paragraph 6.3.

3.9 Mechanical properties.

3.9.1 Type I foam. The minimal mechanical properties of type I foam shall be as specified in Table III.

TABLE III. Minimum mechanical properties of type I foam.

Class	Compressive Strength		Tensile				Shear			
			Strength		Modulus		Strength		Modulus	
	psi	(kPa)	psi	(kPa)	psi	(kPa)	psi	(kPa)	psi	(kPa)
1	29	(200)	80	(552)	3.2	(22.1)	42.6	(293.7)	1.1	(7.6)
2	61	(421)	152	(1048)	6.2	(42.7)	72.5	(499.9)	2.0	(13.8)
3	137	(945)	246	(1696)	10.6	(73.1)	127.0	(875.7)	3.0	(20.7)

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3.9.2 Type II foam. The minimal mechanical properties of type II foam shall be as specified in Table IV.

TABLE IV. Minimal mechanical properties of type II foam. [1/]

Class	Compressive Strength		Tensile				Shear			
			Strength		Modulus		Strength		Modulus	
	psi	(kPa)	psi	(kPa)	psi	(kPa)	psi	(kPa)	psi	(kPa)
2	58	(400)	128	(883)	6.5	(44.8)	72	(496)	2.0	(13.8)
3	152	(1048)	227	(1565)	10.8	(74.5)	145	(1000)	3.5	(24.1)
4	319	(2200)	369	(2544)	19.5	(134.4)	253	(1744)	5.8	(40.0)
5	928	(6399)	780	(5378)	39.1	(269.6)	426	(2937)	14.5	(100.0)
6	1150	(7929)	1180	(8136)	43.5	(299.9)	760	(5240)	29.0	(200.0)

[1/] Aerospace users may specify higher values for structural applications in consultation with the manufacturer/supplier.

4. VERIFICATION

4.1 Verification alternatives. Alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures may be proposed by the contractor. Acceptable alternative verification approaches shall be identified in the contract.

4.2 Classification of inspection. The examination and testing of the foam material shall be classified as follows:

a. First article inspection (see 4.2.1)
b. Conformance inspection (see 4.2.2)

4.2.1 First article inspection. First article inspection shall be performed on the first production-representative samples of an order when a first article sample is required (see 3.1). The first article sample(s) shall be examined for all the provisions specified by the procuring activity, the contract or the purchase order (see 6.2). This inspection shall consist of the tests specified in table V.

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TABLE V. Test Methods.

Property	Test Method
Compressive strength	ASTM D1621 1/, 2/
Apparent density	ASTM D1622 3/
Shear strength	ASTM C273 4/
Shear modulus	ASTM C273 4/
Tensile strength	ASTM D638 5/
Tensile modulus	ASTM D638 6/

1/ Procedure A in ASTM D1621.
2/ Specimen size: 2 in. x 2 in. x 1 in. (5 x 5 x 2.5 cm), speed 0.2 in./min (0.5 cm/min).
3/ Specimen size: 4 in. x 4 in. x 1 in. (10 x 10 x 2.5 cm).
4/ Compression method, speed: 0.08 in./min (0.20 cm/min), thickness: 0.78 in. (1.98 cm).
5/ Specimen type I, 0.4 in. x 0.4 in. (1 x 1 cm), speed: 0.2 in/min (0.5 cm/min).
6/ Specimen type I, 0.4 in. x 0.4 in. (1 x 1 cm), speed: 0.04 in/min (1 mm/min).

4.2.2 Conformance inspection. Conformance inspection for acceptance of rigid, structural, closed cell foam shall meet all the provisions specified by the procuring activity, the contract or the purchase order (see 6.2).

4.3 Lot size. Lot size shall consist of all foam material of the same density produced from one batch during a single production run.

4.4 Sampling. Unless otherwise specified, at least one sample sheet which is representative of each lot (4.3) shall be selected at random for conformance testing. When material is supplied in thicknesses of less than 1 inch (2.5 cm), the supplier shall also furnish material from the same lot, not less than 1 inch (2.5 cm) thick for conformance inspection.

4.5 Water absorption test. Water absorption shall be determined on five specimens selected at random from each density submitted for testing. Each specimen shall be 2 inches by 2 inches by 1 inch thick (5 x 5 x 2.5 cm). The specimens shall not be sealed or protected with any coating which would inhibit moisture absorption. After drying as specified in 6.3, the specimens shall be allowed to cool to room temperature (73.4 ± 3.6°F) (23.0 ± 2.0°C) in an environmentally controlled (50 ± 5% RH) room for 2 hours and then shall be weighed. The specimens shall then be placed into an environmental chamber at a relative humidity of 85 ± 2 percent and a temperature of 160 ± 5°F (71 ± 3°C).

Specimens shall be removed from the chamber periodically (but at least once a week), allowed to cool to room temperature, then reweighed to determine weight after exposure. The above procedure shall be repeated until the equilibrium moisture level (saturation) is attained. Equilibrium shall be defined as two consecutive weekly measurements indicating a moisture gain

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of less than 0.05 percent. Except for reweighing, wet conditioning shall not be interrupted. Calculate moisture absorption for each specimen as:

$$\text{weight (percent) absorbed} = \frac{(\text{weight wet specimen - weight dry specimen})}{(\text{weight dry specimen})} \times 100$$

Then calculate the average moisture absorbed.

4.6 Noncompliance.

4.6.1 Rejection. Unless otherwise specified where one or more test specimens fail to meet the requirements of the specification, the lot represented by the specimen or specimens shall be subject to rejection.

4.6.2 Retest. When a sampling plan is not provided or approved by the procuring agency, where there is evidence that indicates that the specimen was not representative of the lot of material, and when the contract does not otherwise specify, at least two specimens shall be selected to replace each test specimen which failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The foams which are specified in this document are rigid, closed cell materials comparable in density to some types of honeycomb core. This material may be a viable candidate for structural aerospace applications whenever a core or filler is needed and in nonstructural applications including formed configurations. Polymethacrylimide (PMI) foam has been used to comply with the requirements of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification

b. Type and class of material (see 1.2)

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c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

d. When first article is required (see 3.1).

e. Dimensions in SI and English units.

f. Sampling plan, if other than specified in 4.4

g. Packaging requirements (see 5.1).

6.3 Procedure for drying type I and type II foams. Dry foam sections and panels in an air circulating oven at 250 ± 10°F (121 ± 5°C) for a minimum of 2 hours. Do not allow panels to come in contact with other panels in the oven (do not stack panels). Separate each panel from the closest panel by not less than one inch (2.5 cm). Arrange panels parallel to the direction of air flow. Do not place panels directly on the bottom of the oven, or on any other piece of nonperforated metal (which could restrict air flow). Place a piece of perforated metal or heavy metal screen atop the panel to prevent warpage during the drying cycle.

6.4 Procedure for heat treating type II foams. Prior to heat treating, dry the material in accordance with 6.3. Type II foams should then be heat treated in accordance with instructions provided by the manufacturer. When no manufacturer instructions exist or are available, end users should heat treat the material immediately after the drying cycle for the following times and temperatures: 48 hours @ 350 ± 10°F (177 ± 5°C) for class 2 and 3, 20 hours @ 320 ± 10°F (160 ± 5°C) followed by 28 hours @ 350 ± 10°F (177 ± 5°C) for class 4 and 5, and 24 hours @ 350 ± 10°F (177 ± 5°C) for class 6. Do not allow panels to come in contact with other panels in the oven (do not stack panels) and separate each panel from the closest panel by not less than one inch (2.5 cm). Arrange panels parallel to the direction of air flow. Do not place panels directly on the bottom of the oven, or on any other piece of nonperforated metal (which could restrict air flow). Place a piece of perforated metal or heavy metal screen on both sides of each panel to prevent warpage during the heat treating cycle. Heat treating is a processing step involving extended exposure of foam to elevated temperatures; heat treat under highly controlled conditions to ensure dimensional stability of the foam.

6.5 Subject term (key word) listing.

Honeycomb core	Polymethacrylimide (PMI)

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS
Air Force - 11

Preparing activity:
ARMY - MR

(Project CMPS-2025-001)

Review activities:
Army - AV, EA, MI, SM
Navy - SH
DLA - GS

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